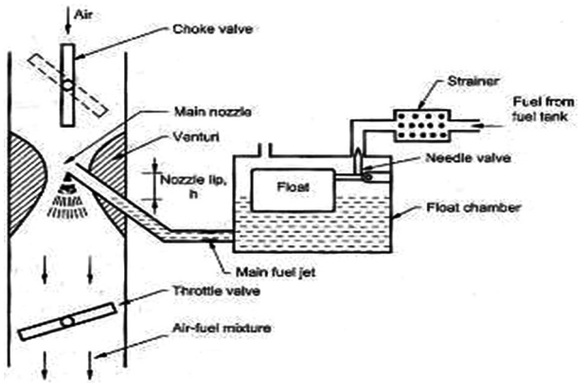


Figure 9.5.4 Simple Carburettor

fuel (petrol) ratio correctly according to the load and speed of the engine in a carburettor works it can be called as simple carburettor. Figure 9.5.4 shows the Simple Carburettor.

The parts of simple carburettor

1) Float chamber
2) venture (or) choke tube
3) main jet
4) choke valve
5) throttle valve

Working:

In float chamber there be a float and a needle valve, usually the float will be thin copper plate or plastic material of less weight and it will be made such that there will be Vacuum inside.

The inlet petrol from the fuel pump will be attached to the top position of the float chamber. So that when the petrol comes into the float chamber when it reaches the certain level the float moves upwards. When the float chamber is full the need valve in the float block the petrol line. Like the same way when the petrol level reaches down the float comes down and allows petrol to come in. so that the level is maintained inside the chamber always.

The level of the petrol in the float chamber will be 5mm less than needle height. So that when the engine at rest and while climbing hills it won't allow petrol to over flow in the mixing chamber.

To control the amount of air fuel mixture from the carburettor to the engine, there will be throttle valve in the carburettor. This valve will be connected to the accelerator cable with suitable arrangements. So that when the driver presses the accelerator pedal the amount of air fuel mixture goes inside the engine accordingly.

During the suction stroke of the engine more air will be sucked inside via air filter. In the air flow passage there will be venture with convergent type. When the air crosses the venture the pressure reduces and speed is increased around the main nozzle. So that when the petrol enters the nozzle with less pressure it can be vaporized easily and mixes with the air. It crosses the throttle valve inlet manifold and inlet valve and reaches the engine cylinder.

In cold climate when engine is started the choke valve is being closed using the choke cable. So that the air flow passage is closed for some time at that time the petrol and less amount of air in the mixing chamber and enters the nozzle as rich mixture. So that it will be easy to start engine as petrol high and less air (rich mixture).

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